

FLORIANOPOLIS, Brazil

WMO: 838990

Lat: 27.670S

Lon: 48.547W

Elev: 20

StdP: 14.69

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	46.7	50.0	37.7	33.3	54.4	41.2	38.2	55.0	19.5	64.1	17.7	63.4	4.0	300	0.332

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.4	89.9	77.6	87.9	77.1	86.0	76.1	79.7	86.3	78.6	84.8	77.7	83.4	10.2	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.5	143.2	82.3	77.0	140.6	81.7	75.6	134.1	80.4	43.3	86.7	42.2	85.5	41.1	83.2	86.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.7	15.1	DB	41.0	96.2	3.8	3.2	38.3	98.5	36.1	100.4	33.9	102.1	31.2	104.5	(4)
(5)			WB	39.1	81.9	3.2	1.3	36.8	82.9	34.9	83.7	33.2	84.4	30.8	85.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	70.6	77.8	78.0	76.3	72.8	67.6	64.3	62.7	64.4	66.1	69.5	72.1	75.9	(6)	
	DBStd	7.05	3.72	3.77	3.81	4.41	4.64	5.38	5.71	5.55	4.73	4.10	4.07	4.23	(7)	
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD65	351	0	0	0	3	27	77	112	81	42	8	1	0	(9)	
	CDD50	7512	863	785	816	683	544	429	393	448	483	603	663	802	(10)	
	CDD65	2388	398	365	351	236	106	56	40	64	75	146	214	337	(11)	
	CDH74	13047	2917	2700	2229	1079	272	137	104	170	157	367	855	2060	(12)	
	CDH80	3058	783	731	534	189	33	16	18	37	18	43	140	516	(13)	
(14) Wind	WSAvg	7.2	7.2	6.7	6.4	6.3	6.2	6.5	6.7	7.9	8.2	8.1	8.3	7.9	(14)	
(15) Precipitation	PrecAvg	63.1	8.7	8.1	6.7	4.7	4.0	3.0	3.5	3.1	5.7	5.8	5.4	5.7	(15)	
	PrecMax	93.0	24.6	21.1	12.8	12.4	12.2	9.6	7.3	10.3	12.1	13.3	23.4	13.4	(16)	
	PrecMin	32.6	1.1	2.1	2.0	1.3	0.1	0.3	0.2	0.4	0.7	1.4	1.4	0.4	(17)	
	PrecStd	14.4	4.3	4.2	2.8	2.5	2.6	1.8	1.8	2.2	2.8	2.7	3.7	2.9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.1	93.2	91.3	87.6	84.0	82.3	82.5	84.9	82.2	84.2	87.8	92.6	(19)	
		MCWB	79.2	79.1	78.5	76.5	72.7	69.7	68.2	68.9	70.8	73.3	75.4	78.2	(20)	
	2%	DB	89.4	89.6	87.8	84.5	79.2	77.4	76.7	78.2	78.3	80.6	83.8	87.9	(21)	
		MCWB	77.8	77.9	77.6	74.6	70.4	68.8	67.2	68.2	69.1	71.2	73.2	76.6	(22)	
	5%	DB	86.4	87.0	85.6	82.2	76.9	74.8	73.3	75.0	75.2	77.9	80.8	85.2	(23)	
		MCWB	76.8	77.3	76.4	73.3	69.4	67.2	65.9	66.8	67.7	70.0	71.6	75.3	(24)	
	10%	DB	84.3	84.3	82.7	79.4	74.8	71.8	70.1	71.8	72.4	75.4	78.8	82.5	(25)	
		MCWB	75.9	76.1	74.7	72.3	68.3	65.9	64.4	65.2	66.2	68.7	70.5	74.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.0	80.7	80.4	78.3	74.6	72.2	70.1	70.9	72.0	75.0	77.0	80.1	(27)	
		MCDB	87.6	87.9	86.9	84.2	80.6	78.4	77.9	79.5	79.2	81.7	84.6	87.6	(28)	
	2%	WB	79.5	79.3	78.9	76.4	72.5	70.0	68.1	69.0	70.1	72.9	74.9	78.3	(29)	
		MCDB	85.9	85.9	84.7	81.6	77.1	75.0	74.1	76.0	75.1	77.5	80.6	84.6	(30)	
	5%	WB	78.3	78.3	77.7	75.2	70.9	68.5	66.7	67.5	69.0	71.5	73.7	76.8	(31)	
		MCDB	84.2	84.3	83.0	79.8	75.1	73.1	71.6	73.0	73.2	75.9	78.3	82.6	(32)	
	10%	WB	77.3	77.3	76.4	74.1	69.4	67.1	65.4	66.5	67.7	70.2	72.4	75.6	(33)	
		MCDB	82.5	82.7	81.2	78.0	73.3	71.2	69.4	71.1	71.2	73.9	76.8	80.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.4	12.5	12.9	13.2	13.3	13.4	13.5	11.8	11.2	12.2	12.4	(35)	
		MCDBR	15.3	15.4	14.7	15.0	15.2	15.7	16.3	16.3	14.6	14.1	14.9	15.8	(36)	
	5% WB	MCWBR	6.7	6.7	6.6	7.1	7.5	7.7	8.4	7.7	7.4	7.2	7.2	7.2	(37)	
		MCWBR	14.0	14.1	13.7	12.8	13.4	14.4	14.8	14.8	12.9	12.4	13.3	14.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.418	0.413	0.400	0.397	0.370	0.391	0.399	0.469	0.535	0.466	0.424	0.420	(40)	
		taud	2.438	2.452	2.481	2.464	2.512	2.441	2.389	2.138	1.952	2.217	2.366	2.405	(41)	
	Ebn at Noon	Ebn at Noon	294	290	284	269	262	247	249	241	240	271	290	294	(42)	
		Edh at Noon	39	37	35	32	28	29	31	43	57	47	42	40	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1864	1750	1524	1240	959	830	892	1074	1223	1419	1745	1860	(44)	
	RadStd	RadStd	103	141	126	113	102	57	99	112	135	183	163	151	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page